Chapter 3. Continuous Functions

Limits and continuity, extreme and intermediate values, uniform continuity, monotone functions

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Throughout, $X \subset \mathbb{R}^n$, $Y \subset \mathbb{R}^m$, and functions $f : X \to Y$.

3.1 Limits and continuity

Motivation. Continuity means small changes in input produce small changes in output. We define it via limits and via sequences, and show the two agree, so that the sequence results of Chapters 1–2 apply directly to functions.

Let x_0 be a limit point of X. We write $\lim_{x\to x_0} f(x) = y_0$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - y_0\| < \epsilon$ whenever $x \in X$ and $0 < \|x - x_0\| < \delta$. f is **continuous at** $x_0 \in X$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(x_0)\| < \epsilon$ whenever $x \in X$, $\|x - x_0\| < \delta$; equivalently, x_0 is isolated in X, or $\lim_{x\to x_0} f(x) = f(x_0)$. f is **continuous** if it is continuous at every point.

Proposition 3.1 (3.1.1)

If $\lim_{x\to x_0} f(x) = y_0$ and $\lim_{x\to x_0} g(x) = z_0$, then $f + g \to y_0 + z_0$, $af \to ay_0$, and for real-valued functions $fg \to y_0 z_0$, and $f/g \to y_0/z_0$ if $z_0 \neq 0$.

Proof. As in Theorem 1.14, or by Proposition 3.3 below applied to sequences. ■

Proposition 3.2 (3.1.2)

If $f, g : X \to \mathbb{R}^m$ are continuous, so is $af + bg$; for real-valued f, g, so are fg and f/g (where $g \neq 0$).

Proof. Apply Proposition 3.1 at each non-isolated point. ■

Polynomials in several variables are therefore continuous, and so are rational functions where the denominator does not vanish.

Proposition 3.3 (3.1.3, sequential criterion)

f is continuous at x_0 iff $f(x_k) \to f(x_0)$ for every sequence $x_k \to x_0$ in X.

Proof. ($\Rightarrow$) Given ϵ, take δ; for large k, $\|x_k - x_0\| < \delta$, so $\|f(x_k) - f(x_0)\| < \epsilon$.

($\Leftarrow$) If f is not continuous at x_0, for some ϵ and each k there is x_k with $\|x_k - x_0\| < 1/k$ but $\|f(x_k) - f(x_0)\| \geq \epsilon$. ■

Similarly $\lim_{x\to x_0} f(x) = y_0$ iff $f(x_k) \to y_0$ for every sequence $x_k \to x_0$ with $x_k \neq x_0$.

Proposition 3.4 (3.1.4)

If $f : X \to Y$ is continuous at x_0 and $g : Y \to Z$ is continuous at $f(x_0)$, then $g \circ f$ is continuous at x_0.

Proof. Given ϵ, choose η for g at $f(x_0)$ and then δ for f at x_0 with η in place of ϵ. ■

Example 3.1

(a) $f(x) = \sin(1/x)$ $(x \neq 0)$ has no limit at 0: $x_k = 1/(k\pi + \pi/2)$ gives ± 1 alternately.
(b) $x \sin(1/x) \to 0$ by the squeeze $|x \sin(1/x)| \leq |x|$.
(c) Thomae's function ($f(p/q) = 1/q$ in lowest terms, $f = 0$ at irrationals) is continuous

exactly at the irrationals.
(d) $f(x,y) = xy/(x^2+y^2)$, $f(0,0) = 0$, is continuous along every line through the origin, but $f(t,t) = \frac{1}{2}$, so f is not continuous at 0.

Theorem 3.5 (3.1.5)

For $f : X \to Y$ the following are equivalent:
(a) f is continuous;
(b) $f^{-1}(V)$ is open in X for every open V in Y;
(c) $f^{-1}(E)$ is closed in X for every closed E in Y.

Proof. (a)⇒(b): if $x \in f^{-1}(V)$ and $N(f(x),\epsilon) \cap Y \subset V$, the δ of continuity gives $N(x,\delta) \cap X \subset f^{-1}(V)$. (b)⇒(a): $f^{-1}(N(f(x_0),\epsilon) \cap Y)$ is open in X and contains x_0. (b)⇔(c): $f^{-1}(Y - E) = X - f^{-1}(E)$. ■

3.2 Continuity, compactness and connectedness

Motivation. The classical theorems about continuous functions on $[a,b]$, that they attain a maximum and take every intermediate value, are really statements that continuity preserves compactness and connectedness. Proving them this way makes them hold in $\mathbb{R}^n$ as well.

Theorem 3.6 (3.2.1)

If $f : X \to Y$ is continuous and surjective and X is compact, then Y is compact.

Proof. Let $\{U_i\}$ be an open cover of Y; by Theorem 3.5 and Proposition 2.6, $f^{-1}(U_i) = V_i \cap X$ with V_i open in $\mathbb{R}^n$, and the V_i cover X. Finitely many $V_{i_1}, \ldots, V_{i_k}$ do; then $U_{i_1}, \ldots, U_{i_k}$ cover $Y = f(X)$. ■

Corollary 3.7 (3.2.2, extreme value theorem)

A continuous $f : X \to \mathbb{R}$ on a compact set attains its maximum and minimum.

Proof. $f(X)$ is compact, hence bounded and closed; so $\sup f(X)$ is finite and, being a limit of points of $f(X)$, lies in $f(X)$. ■

Corollary 3.8 (3.2.3)

If X is compact and $f : X \to \mathbb{R}$ is continuous with $f > 0$, then there is $\delta > 0$ with $f(x) \geq \delta$ for all x.

Proof. Take $\delta = \min f > 0$. ■

Corollary 3.9 (3.2.4)

A continuous $f : [a,b] \to \mathbb{R}$ attains its maximum and minimum.

Corollary 3.10 (3.2.5)

If X is compact and $f : X \to Y$ is a continuous bijection, then f^{-1} is continuous.

Proof. Let U be open in X. $X - U$ is closed in the compact X, so compact (Proposition 2.24); then $f(X - U) = Y - f(U)$ is compact, hence closed, so $f(U)$ is open in Y. By Theorem 3.5, f^{-1} is continuous. ■

Compactness is needed: $t \mapsto (\cos t, \sin t)$ is a continuous bijection $[0, 2\pi) \to S^1$ whose inverse is not continuous. For nonempty $A, B \subset \mathbb{R}^n$ let $\operatorname{dist}(A, B) = \inf\{\|a - b\| : a \in A, b \in B\}$.

Proposition 3.11 (3.2.6)

If A is compact, B is closed and $A \cap B = \emptyset$, then $\operatorname{dist}(A, B) > 0$.

Proof. The function $d(x) = \operatorname{dist}(\{x\}, B)$ is continuous (indeed $|d(x) - d(y)| \le \|x - y\|$) and positive on A, since $\mathbb{R}^n - B$ is open. By Corollary 3.8 it is bounded below on A by some $\delta > 0$. ■

The compactness is needed: the closed sets $\{(x, 0)\}$ and $\{(x, 1/x) : x > 0\}$ are at distance 0.

Lemma 3.12 (3.2.7)

S is disconnected iff there is a continuous surjection $f : S \to \{0, 1\}$.

Proof. If U, V disconnect S, let $f = 0$ on $U \cap S$ and 1 on $V \cap S$; preimages of subsets of $\{0, 1\}$ are relatively open, so f is continuous. Conversely, $f^{-1}(0)$ and $f^{-1}(1)$ are nonempty and relatively open (Theorem 3.5), so they are $U \cap S$ and $V \cap S$ with U, V open (Proposition 2.6); then $U \cap V \cap S = f^{-1}(0) \cap f^{-1}(1) = \emptyset$ and $S \subset U \cup V$. ■

Theorem 3.13 (3.2.8)

If $f : X \to Y$ is continuous and surjective and X is connected, then Y is connected.

Proof. If Y were disconnected, Lemma 3.12 gives a continuous surjection $g : Y \to \{0, 1\}$, and then $g \circ f : X \to \{0, 1\}$ is a continuous surjection. ■

Corollary 3.14 (3.2.9)

If $I \subset \mathbb{R}$ is an interval and $f : I \to \mathbb{R}$ is continuous, then $f(I)$ is an interval.

Proof. Proposition 2.28 and Theorem 3.13. ■

Corollary 3.15 (3.2.10, intermediate value theorem)

If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < \alpha < f(b)$, then $f(c) = \alpha$ for some $c \in (a, b)$.

Corollary 3.16 (3.2.11)

If $f : [a, b] \to \mathbb{R}$ is continuous, then $f([a, b])$ is a bounded closed interval $[m, M]$.

Proof. It is an interval (Corollary 3.14) and compact (Theorem 3.6), containing its minimum m and maximum M. ■

A function mapping intervals to intervals need not be continuous, for example $\sin(1/x)$ extended by 0 at 0.

Corollary 3.17 (3.2.12, fixed point)

Every continuous $f : [a, b] \to [a, b]$ has a fixed point.

Proof. $g(x) = x - f(x)$ has $g(a) \le 0 \le g(b)$; apply Corollary 3.15 (or $g(a) = 0$ or $g(b) = 0$). ■

Example 3.2

Every polynomial of odd degree has a real root (it takes both signs). Every $a > 0$ has a unique positive n-th root, since x^n is continuous, strictly increasing and unbounded on $[0, \infty)$.

3.3 Uniform continuity

Motivation. Ordinary continuity allows the required δ to depend on the point. When one δ works everywhere we can approximate a function uniformly by step functions or polynomials, which is what integration needs; on compact sets this always happens.

$f : X \to Y$ is **uniformly continuous** if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(y)\| < \epsilon$ whenever $x, y \in X$, $\|x - y\| < \delta$.

Theorem 3.18 (3.3.1, Heine)

A continuous function on a compact set is uniformly continuous.

Proof. Given ϵ, choose for each $x \in X$ a $\delta(x) > 0$ with $\|f(y)-f(x)\| < \epsilon/2$ for $y \in X$, $\|y-x\| < \delta(x)$. The sets $N(x, \delta(x)/2)$ cover X; finitely many with centres $x_1, \dots, x_k$ do. Let $\delta = \min \delta(x_i)/2$. If $\|x - y\| < \delta$, then $\|x - x_i\| < \delta(x_i)/2$ for some i, and $\|y - x_i\| < \delta(x_i)$; so $\|f(x) - f(y)\| \le \|f(x) - f(x_i)\| + \|f(x_i) - f(y)\| < \epsilon$. ■

Example 3.3

x^2 is not uniformly continuous on $\mathbb{R}$ ($(n+1/n)^2 - n^2 > 2$), and $1/x$ is not uniformly continuous on $(0, 1)$; $\sqrt{x}$ is uniformly continuous on $[0, \infty)$ since $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$.

Theorem 3.19 (3.3.2)

A uniformly continuous function maps Cauchy sequences to Cauchy sequences.

Proof. Given ϵ, take δ from uniform continuity and N with $\|x_m - x_n\| < \delta$ for $m, n \ge N$. ■

Corollary 3.20 (3.3.3)

A uniformly continuous $f : X \to \mathbb{R}^m$ extends uniquely to a continuous function on $\overline{X}$, which is again uniformly continuous.

Proof. For $x_0 \in \overline{X}$ pick $x_k \to x_0$ in X; $(f(x_k))$ is Cauchy (Theorem 3.19), hence converges (Theorem 2.14); define $F(x_0)$ as the limit. Interlacing two such sequences shows the limit does not depend on the sequence, and $F = f$ on X. If $\|x - y\| < \delta$ in $\overline{X}$, approximate by $x_k \to x$,

$y_k \to y$ with $\|x_k - y_k\| < \delta$; then $\|F(x) - F(y)\| = \lim \|f(x_k) - f(y_k)\| \le \epsilon$. Uniqueness follows from Proposition 3.3. ■

3.4 Monotone functions

Motivation. Monotone functions are the simplest functions that need not be continuous. Their discontinuities are only jumps and are countable, which makes them well suited to integration and to describing distribution functions.

For $f : (a, b) \to \mathbb{R}$ write $f(x+) = \lim_{t \to x^+} f(t)$ and $f(x-) = \lim_{t \to x^-} f(t)$ when these exist.

Theorem 3.21 (3.4.1)

A monotone $f : (a, b) \to \mathbb{R}$ has one-sided limits at every point, and $f(a+)$, $f(b-)$ exist in $[-\infty, \infty]$. If f is increasing and $a < x < y < b$, then $f(x-) \le f(x) \le f(x+) \le f(y-) \le f(y) \le f(y+)$, and $f(x+) = \inf_{t>x} f(t)$, $f(x-) = \sup_{t<x} f(t)$.

Proof. Let f increase. $A = \{f(t) : a < t < x\}$ is bounded above by $f(x)$; let $s = \sup A$. Given ϵ, some $f(t_0) > s - \epsilon$, and for $t_0 < t < x$, $s - \epsilon < f(t_0) \le f(t) \le s$; so $f(x-) = s \le f(x)$. Similarly $f(x+) = \inf_{t>x} f(t) \ge f(x)$. For $x < y$, choosing $x < t < y$, $f(x+) \le f(t) \le f(y-)$. ■

Corollary 3.22 (3.4.2)

The set of discontinuities of a monotone function on an interval is countable.

Proof. For increasing f, f is discontinuous at x iff $f(x-) < f(x+)$. Choose a rational $r(x) \in (f(x-), f(x+))$; by Theorem 3.21 these open intervals are disjoint for distinct x, so $x \mapsto r(x)$ is injective into $\mathbb{Q}$. ■

Conversely, for any countable set $\{x_k\} \subset (a, b)$, $f(x) = \sum_{x_k < x} 2^{-k}$ is increasing and discontinuous exactly at the x_k.

Theorem 3.23 (3.4.3)

For a continuous $f : I \to \mathbb{R}$ on an interval, f is injective iff it is strictly monotone. In that case $f(I)$ is an interval and $f^{-1} : f(I) \to I$ is continuous.

Proof. Strictly monotone functions are injective. Conversely, suppose f injective but not strictly monotone; then there are $x < y < z$ in I with $f(y)$ not between $f(x)$ and $f(z)$, say $f(y) > \max(f(x), f(z))$. A value α with $\max(f(x), f(z)) < \alpha < f(y)$ is attained in (x, y) and in (y, z) (Corollary 3.15), contradicting injectivity. If f is strictly increasing, f^{-1} is strictly increasing on the interval $f(I)$ (Corollary 3.14) and maps it onto the interval I; a monotone function with an interval as image has no jumps (a jump $g(y-) < g(y+)$ would omit values by Theorem 3.21), so f^{-1} is continuous. ■

3.5 Selected exercises

Motivation. Four problems from the book's exercise set. Each turns one tool of this chapter (density plus the sequential criterion, compactness, connectedness, completeness) into a concrete statement about continuous functions.

By Proposition 2.3(b) and Theorem 2.4, a set $F \subset \mathbb{R}^n$ is closed iff the limit of every convergent sequence in F lies in F; we use this repeatedly.

Exercise 3.5.2 (Cauchy's functional equation)

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous with $f(x+y) = f(x) + f(y)$ for all x, y. Show that $f(x) = ax$ for all x, where $a = f(1)$.

Solution.

Step 1. $f(0) = f(0+0) = 2f(0)$, so $f(0) = 0$; then $0 = f(r + (-r)) = f(r) + f(-r)$ gives $f(-r) = -f(r)$.

Step 2. By induction on $n \in \mathbb{N}$, $f(nr) = nf(r)$: it holds for $n = 0$, and $f((n+1)r) = f(nr)+f(r) = (n+1)f(r)$. With Step 1 it holds for all $n \in \mathbb{Z}$.

Step 3. For $n \geq 1$, $a = f(1) = f(n \cdot \frac{1}{n}) = nf(\frac{1}{n})$, so $f(\frac{1}{n}) = \frac{a}{n}$, and for $m \in \mathbb{Z}$, $f(\frac{m}{n}) = mf(\frac{1}{n}) = a\frac{m}{n}$. Thus $f(q) = aq$ for every $q \in \mathbb{Q}$.

Step 4. Let $x \in \mathbb{R}$. By Theorem 1.8 choose $q_k \in \mathbb{Q}$ with $x < q_k < x + 1/k$, so $q_k \to x$. By Proposition 3.3, $f(x) = \lim f(q_k) = \lim aq_k = ax$. ■

Exercise 3.5.4 (Closed graphs)

For $f : \mathbb{R} \to \mathbb{R}$ let $G = \{(x, f(x)) : x \in \mathbb{R}\} \subset \mathbb{R}^2$ be its graph.
(a) Show that if f is continuous, then G is closed.
(b) Show that if f is bounded and G is closed, then f is continuous.
(c) Show by an example that boundedness cannot be dropped in (b).

Solution.

(a) Let $(x_k, f(x_k)) \to (x, y)$. Then $x_k \to x$ and $f(x_k) \to y$ (coordinatewise convergence). By Proposition 3.3, $f(x_k) \to f(x)$, so $y = f(x)$ and $(x, y) \in G$. Hence G is closed.

(b)

Step 1. Let $|f| \leq M$ and suppose f is not continuous at x. By Proposition 3.3 there are $\epsilon > 0$ and $x_k \to x$ with $|f(x_k) - f(x)| \geq \epsilon$ for all k.

Step 2. $(f(x_k))$ is bounded, so a subsequence $f(x_{k_j})$ converges to some y (Theorem 2.10), and $|y - f(x)| \geq \epsilon$.

Step 3. Then $(x_{k_j}, f(x_{k_j})) \to (x, y)$, and since G is closed, $(x, y) \in G$, i.e. $y = f(x)$. This contradicts Step 2.

(c) Let $f(x) = 1/x$ for $x \neq 0$ and $f(0) = 0$; f is not continuous at 0. Its graph is closed: let $(x_k, f(x_k)) \to (x, y)$.

Step 1. If $x \neq 0$, then $x_k \neq 0$ for large k and $f(x_k) = 1/x_k \to 1/x$, so $y = f(x)$.

Step 2. If $x = 0$: the convergent sequence $(f(x_k))$ is bounded, say by $K > 0$. Whenever $x_k \neq 0$, $1/|x_k| = |f(x_k)| \leq K$, i.e. $|x_k| \geq 1/K$. Since $x_k \to 0$, this forces $x_k = 0$ for all large k, and $y = \lim f(x_k) = 0 = f(0)$. ■

Exercise 3.5.9 (Antipodal points with equal values)

Let $T = \{(\cos t, \sin t) : t \in \mathbb{R}\} \subset \mathbb{R}^2$ be the unit circle and $f : T \to \mathbb{R}$ continuous. Show that $f(x) = f(-x)$ for some $x \in T$.

Solution.

Step 1. T is the image of the interval $[0, 2\pi]$ under the continuous map $t \mapsto (\cos t, \sin t)$, so T is connected by Proposition 2.28 and Theorem 3.13.

Step 2. $x \mapsto -x$ maps T into T and is continuous, so $g(x) = f(x) - f(-x)$ is continuous on T (Propositions 3.2 and 3.4). By Theorem 3.13 and Proposition 2.28, $g(T)$ is an interval.

Step 3. With $p = (1, 0)$ we have $g(-p) = f(-p) - f(p) = -g(p)$. So the interval $g(T)$ contains $g(p)$ and $-g(p)$, hence 0. A point x with $g(x) = 0$ satisfies $f(x) = f(-x)$.

For example, at any moment there are two antipodal points on the equator with the same temperature. ■

Exercise 3.5.10 (Contraction mapping principle)

Let $X \subset \mathbb{R}^n$ be nonempty and closed, $0 < r < 1$, and $f : X \to X$ with $\|f(x)-f(y)\| \le r\|x-y\|$ for all $x, y \in X$.
(a) Show that f is uniformly continuous.
(b) For $x_0 \in X$ let $x_{k+1} = f(x_k)$. Show that (x_k) is a Cauchy sequence.
(c) Show that f has exactly one fixed point x, that $x_k \to x$, and that $\|x_k - x\| \le \frac{r^k}{1-r}\|x_1 - x_0\|$.

Solution.

(a) Given $\epsilon > 0$ take $\delta = \epsilon$: if $\|x - y\| < \delta$, then $\|f(x) - f(y)\| \le r\|x - y\| < \epsilon$.

(b)

Step 1. By induction, $\|x_{j+1} - x_j\| = \|f(x_j) - f(x_{j-1})\| \le r\|x_j - x_{j-1}\| \le \cdots \le r^j\|x_1 - x_0\|$.

Step 2. For $m > k$, by the triangle inequality,

$$\|x_m - x_k\| \le \sum_{j=k}^{m-1} \|x_{j+1} - x_j\| \le \|x_1 - x_0\| \sum_{j=k}^{\infty} r^j = \frac{r^k}{1-r}\|x_1 - x_0\|.$$

Step 3. Since $r^k \to 0$ (Example 1.1(b)), the right side is $< \epsilon$ for all large k, so (x_k) is Cauchy.

(c)

Step 1. By Theorem 2.14, $x_k \to x$ for some $x \in \mathbb{R}^n$, and $x \in X$ because X is closed.

Step 2. f is continuous by (a), so $f(x) = \lim f(x_k) = \lim x_{k+1} = x$ (Proposition 3.3).

Step 3. If also $f(y) = y$, then $\|x - y\| = \|f(x) - f(y)\| \le r\|x - y\|$, so $(1 - r)\|x - y\| \le 0$ and $y = x$.

Step 4. Letting $m \to \infty$ in Step 2 of (b) gives the error bound $\|x - x_k\| \le \frac{r^k}{1-r}\|x_1 - x_0\|$.

Closedness is needed: $f(x) = x/2$ on $X = (0, 1]$ is a contraction without a fixed point. ■